AMERICAN TERAWATT

HVDC Design

System Topologies

American Terawatt | July 20, 2026

Insulation Coordination

Overvoltages and insulation strength

Insulation coordination

IEEE 1313

“The selection of **insulation strength** consistent with expected **overvoltages** to obtain an acceptable risk of failure.”

Scope of this module:

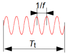
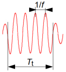
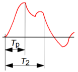
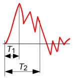
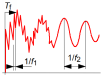
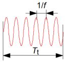
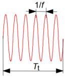
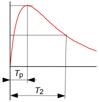
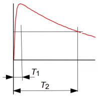
- Overvoltages
- Insulation strength
- Air clearances
- Arresters

Overvoltages

Overvoltages definition according to IEC

IEC 60071-1:2019 © IEC 2019 – 17 –

Table 1 – Classes and shapes of overvoltages, standard voltage shapes and standard withstand voltage tests

Class	Low frequency		Transient		
	Continuous	**Temporary**	**Slow-front**	**Fast-front**	**Very-fast-front**
Voltage or over-voltage shapes					
Range of voltage or over-voltage shapes	f = 50 Hz or 60 Hz $T_t \geq$ 3 600s	10 Hz < f < 500 Hz 0,02 s $\leq T_t \leq$ 3 600 s	20 µs < $T_p \leq$ 5 000 µs $T_2 \leq$ 20 ms	0,1 µs < $T_1 \leq$ 20 µs $T_2 \leq$ 300 µs	$T_f \leq$ 100 ns 0,3 MHz < f_1 < 100 MHz 30 kHz < f_2 < 300 kHz
Standard voltage shapes	 f = 50 Hz or 60 Hz T_t [a]	 48 Hz $\leq f \leq$ 62 Hz T_t = 60 s	 T_p = 250 µs T_2 = 2 500 µs	 T_1 = 1,2 µs T_2 = 50 µs	[a]
Standard withstand voltage test	[a]	Short-duration power frequency test	Switching impulse test	Lightning impulse test	[a]

[a] To be specified by the relevant apparatus committees.

Transient overvoltage in the AC grid

Table 1—Standard withstand voltages for Class I ($15\ kV < V_m \leq 242\ kV$)

Maximum system voltage (phase-to-phase) V_m kV, rms	Low-frequency, short-duration withstand voltage[a] (phase-to-ground) kV, rms	Basic lightning impulse insulation level (phase-to-ground) BIL kV, crest
15	34	95 110
26.2	50	150
36.2	70	200
48.3	95	250
72.5	95 140	250 350
121	140 185 230	350 450 550
145	230 275 325	450 550 650
169	230 275 325	550 650 750
242	275 325 360 395 480	650 750 825 900 975 1050

Table 2—Standard withstand voltages for Class II ($V_m > 242\ kV$)

Maximum system voltage (phase-to-phase) V_m kV, rms	Basic lightning impulse insulation level (phase-to-ground) BIL kV, peak	Basic switching impulse insulation level[a] (phase-to-ground) BSL kV, peak
362	900 975 1050 1175 1300	650 750 825 900 975 1050
550	1300 1425 1550 1675 1800	1175 1300 1425 1550
800	1800 1925 2050	1300 1425 1550 1675 1800

IEEE Std 1313.1-1996

Transient overvoltage in the AC grid

- Overvoltages in the AC grid reflect the “experience of the world”, taking into account protective devices and methods of overvoltage limitation.
- For AC equipment other overvoltage levels can be specified, e.g. transformers.

Table 4—Dielectric insulation levels for all windings of Class II power transformers, voltages in kV

Maximum system voltage (kV rms)	Nominal system voltage[a] (kV rms)	Applied-voltage test[g] (kV rms)			Induced-voltage test[b, c] (phase to ground) (kV rms)		Winding line-end BIL[d] (kV crest)				Neutral BIL[e, g] (kV crest)	
		Delta and fully insulated wye	Grounded wye	Impedance grounded wye or grounded wye with higher BIL	Enhanced 7200 cycle	One hour	Mini-mum	Alternates			Grounded wye	Impedance grounded wye or grounded wye with higher BIL
Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9	Col 10	Col 11	Col 12	Col 13
≤ 17	≤ 15	34	34	34	16	14	**110**				110	110
26	25	50	34	40	26	23	**150**				110	125
36	34.5	70	34	50	36	32	**200**				110	150
48	46	95	34	70	48	42	200	**250**			110	200
73	69	140	34	95	72	63	250	**350**			110	250
121	115	173	34	95	120	105	350	**450**	550		110	250
145	138	207	34	95	145	125	450	**550**	650		110	250
169	161	242	34	140	170	145	550	**650**	750	825	110	350
242	230	345	34	140	240	210	650	**750**	**825**	**900**	110	350
362	345	518	34	140	360	315	900	1050	1175		110	350
550	500	N/A	34	140	550[f]	475[f]	1425	1550	1675		110	350
765	735	N/A	34	140	880[f]	750[f]	1950[f]	2050			110	350
800	765	N/A	34	140	885[f]	795[f]	1950[f]	2050			110	350

IEEE Std C57.12.00™-2021

Temporary overvoltage in the AC grid

- Temporary overvoltages are defined by grid codes such as IEEE 2800, NERC PRC-024 or NERC PRC-029
- Temporary overvoltages do not have an impact on the insulation coordination in the AC grid.

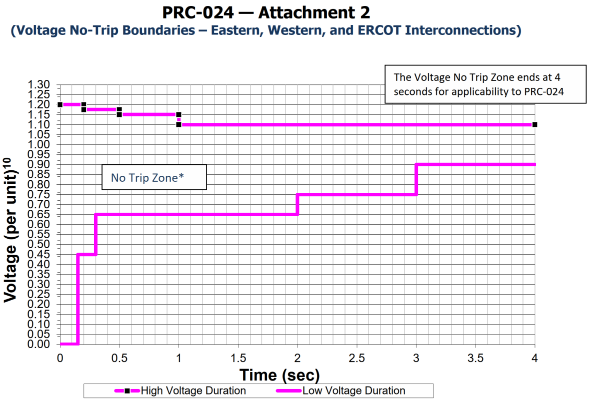


Figure 1

PRC-024-3 - Frequency and Voltage Protection Settings for Generating Resources.

Insulation strength

Insulation strength can be verified by:

- Testing in a high-voltage lab
- Definition of air clearances

www.highvolt.com

Air clearances

Table A.1 – Correlation between standard rated lightning impulse withstand voltages and minimum air clearances

Standard rated lightning impulse withstand voltage	Minimum clearance mm	
kV	Rod-structure	Conductor-structure
20	60	–
40	60	–
60	90	–
75	120	–
95	160	–
125	220	–
145	270	–
170	320	–
200	380	–
250	480	–
325	630	–
380	750	–
450	900	–
550	1 100	–
650	1 300	–
750	1 500	–
850	1 700	1 600
950	1 900	1 700
1 050	2 100	1 900
1 175	2 350	2 200
1 300	2 600	2 400
1 425	2 850	2 600
1 550	3 100	2 900
1 675	3 350	3 100
1 800	3 600	3 300
1 950	3 900	3 600
2 100	4 200	3 900
2 250	4 500	4 150
2 400	4 800	4 450
2 550	5 100	4 700
2 700	5 400	5 000

Table A.2 – Correlation between standard rated switching impulse withstand voltages and minimum phase-to-earth air clearances

Standard rated switching impulse withstand voltage	Minimum phase-to-earth clearance mm	
kV	Rod-structure	Conductor-structure
750	1 900	1 600
850	2 400	1 800
950	2 900	2 200
1 050	3 400	2 600
1 175	4 100	3 100
1 300	4 800	3 600
1 425	5 600	4 200
1 550	6 400	4 900
1 675	7 400	5 600
1 800	8 300	6 300
1 950	9 500	7 200

IEC 60071-1:2019

Environmental conditions

Overvoltages apply for “normal environmental conditions” and “standard reference atmospheric conditions” (IEC 60071-1, §5.9).

Normal environmental conditions

- Temperature range -40 ℃ / $+40$ ℃
- Altitude $< 1000m$
- Ambient air not significantly polluted

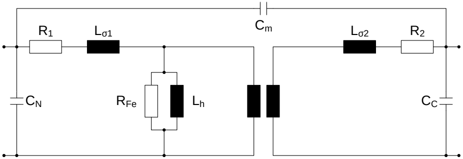
Standard reference atmosphere

- Temperature: 20 ℃
- Pressure: $p_0 = 1013$ hPa
- Absolute humidity: $h_0 = 11$ g/m^3

Outdoors, temperature and absolute humidity compensate each other (no correction required); indoors, correction per IEC 60060-1 is required.

Overvoltage definition

- Overvoltages inside the converter are not based on experience values.
- Definition must be based on EMT-type simulations.
- Component modelling must be adjusted to the anticipated analysis.
- Simulation time step must be appropriate to the simulation.



Transformer representation considering capacitive effects.

Faults for overvoltage definition

- Overvoltages within the converter are defined by converter internal faults.
- Converter internal faults are given by the converter design.
- Fault clearance depends on: DC protection (DC fault detection) and circuit-breaker operation.

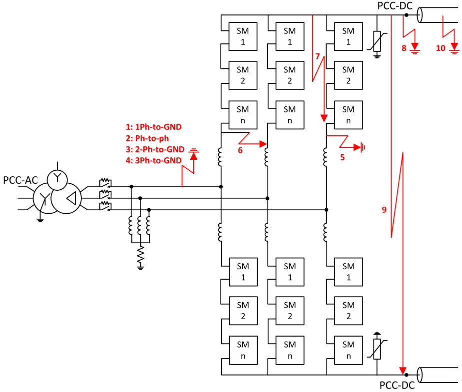
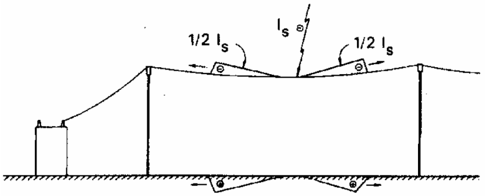


Figure 6-1 Potential fault locations of monopolar system

CIGRE 832 — Guide for Electromagnetic Transient Studies Involving VSC Converters.

Lightning

Lightning stroke



IEEE Std 998-1996

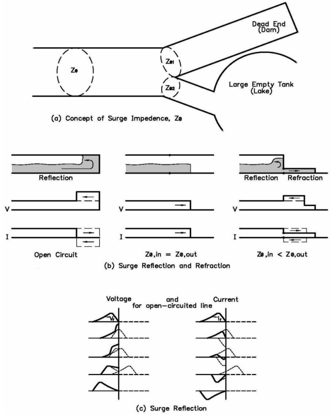
- When a lightning strikes an overhead line, the current wave propagates in two directions
- The travelling current wave creates also a voltage wave travelling along with the current wave
- The voltage wave depends on the lightning current and the surge impedance of the line
- Surge impedance is quite different from DC or 50/60Hz impedances, is not associated with any frequency or line length

Lightning stroke

- Surge impedance depends on the line configuration. Typical values are:
  OHL: $400 - 600\Omega$
  Cable: $20 - 60\Omega$
- The voltage wave travelling with the lightning stroke is defined by:
  $E = \frac{1}{3} \cdot I_S \cdot Z_S$

Travelling wave

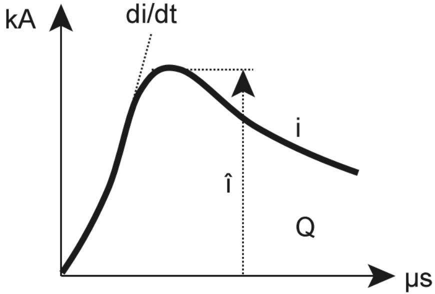
- Travelling-wave propagation is not comparable to the current flow at nominal frequency.
- On an open circuit no current flows, but the travelling wave can still enter the conductor.
- When the wave hits the open circuit (e.g. a transformer) the surge is reflected: the voltage keeps its polarity but the amplitude doubles.
- The increased amplitude at transformers must be considered in the equipment design.



Westinghouse, “Electrical Transmission and Distribution Reference Book”, 1961

Energy

- Energy is defined not only by the peak current but by the total charge Q of the lightning stroke.
- For a $30kA$ stroke and a line surge impedance of 600Ω, the travelling-wave voltage is $\sim 9000kV$.
- Overvoltage causes flashover on line insulators, diverting most of the charge to ground before it reaches protected equipment.
- The most critical strokes hit substations directly — proper lightning protection of equipment must be ensured.



$$Q = \int i \, dt$$
$$i2 = \int i^2 \, dt$$

Arrester

Overvoltages in AC grids

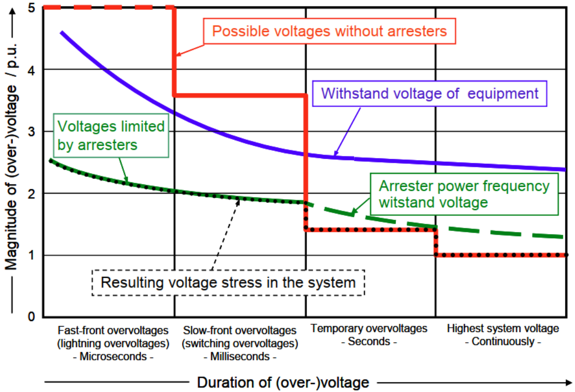


Fig. 1: Schematic representation of the magnitude of voltages and overvoltages in a high-voltage electrical power system versus duration of their appearance (1 p.u. = $\sqrt{2} \cdot U_s / \sqrt{3}$)

“Metal-Oxide Surge Arresters in High-Voltage Power Systems”, Volker Hinrichsen.

Arrester design

- Arresters used today are metal-oxide (MO) arresters without gaps.
- Made from MO resistor disks connected in series.
- The number of series-connected resistors defines the arrester characteristics.
- MO resistors have an extremely non-linear $U - I$ characteristic.
- Resistors are placed in a housing for protection against environmental conditions.

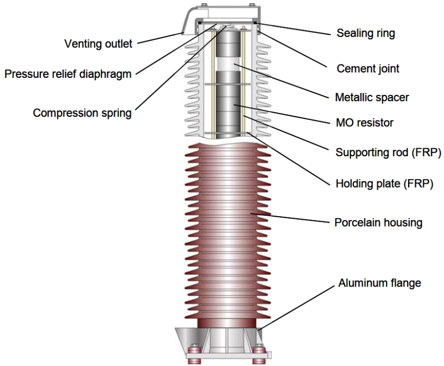


Fig. 8: Cross-sectional drawing of the unit of a porcelain housed MO arrester

"Metal-Oxide Surge Arresters in High-Voltage Power Systems", Volker Hinrichsen

U–I arrester characteristic

- U_S: maximum system voltage (phase-phase AC).
- U_c: maximum continuous operating voltage (MCOV).
- For DC applications the crest value of continuous operating voltage (CCOV) is used instead of MCOV.
- Lightning impulse current (I_n): $8/20\mu s$ (fast-front).
- Switching impulse current: $30/60\mu s$ (slow-front).

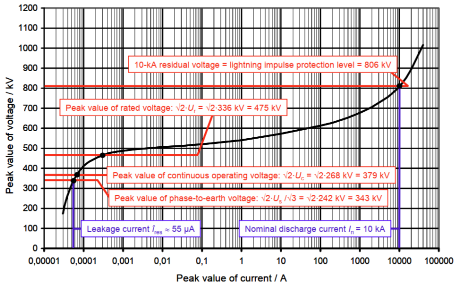


Fig. 2: *U-I*-characteristic of a typical MO arrester in a solidly earthed neutral 420-kV-system

Arrester protection capability — connection

- Protection capability is based on providing a low-impedance path towards earth.
- That impedance is defined by the arrester itself and by its connection to the overhead line and to earth.
- Either accurate calculations or safety margins are applied to account for those effects.

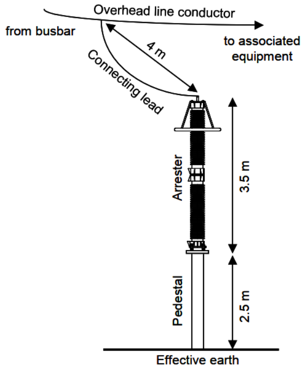


Fig. 6: Typical arrangement of an arrester in a 420-kV substation

Arrester protection capability - travelling wave

- The surge current will enter a “dead end” (e.g. transformer): even with an arrester installed, the travelling wave enters the transformer connection and causes voltage stress.
- The resulting stress is an overlap of the incoming surge, its reflection and the arrester protection rating.
- Equipment voltage stress therefore depends not only on the arrester protective level but also on the distance between arrester and protected device.

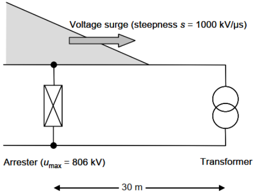


Fig. 5: Simplified arrangement to illustrate the protective zone of an arrester (explanation see text)

“Metal-Oxide Surge Arresters in High-Voltage Power Systems“, Volker Hinrichsen

Energy handling capability

- **Simple impulse handling** — energy injected in only a few µs must not cause mechanical damage.
- **Thermal energy handling** — maximum energy injected during a transient at which the arrester can still cool back to its nominal operating temperature.
- Not precisely defined in the standards, so for VSC-HVDC the energy loading of the arrester is an important design point.

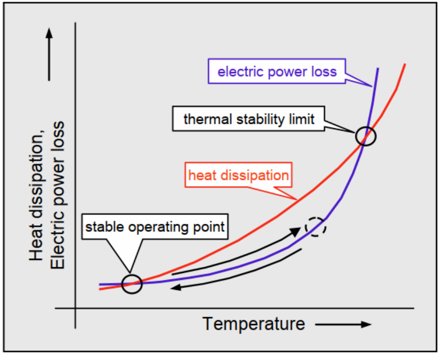
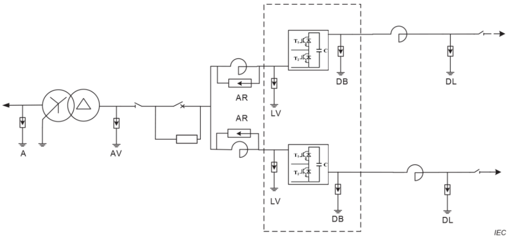


Fig. 7: Explanation of the thermal stability

"Metal-Oxide Surge Arresters in High-Voltage Power Systems", Volker Hinrichsen

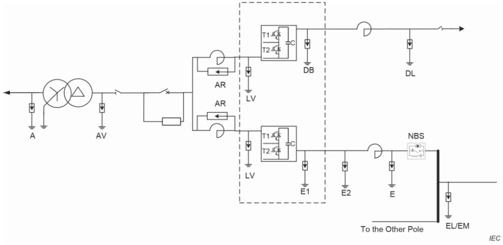
Arrester configuration in VSC-HVDC systems



Key

A:	AC side arrester	LV:	Reactor valve side arrester (bridge arm)
AV:	Connecting transformer valve side arrester	DB:	DC bus arrester
AR:	Arrester between reactor terminals(bridge arm)	DL:	DC line arrester
NV:	Connecting transformer Neutral point arrester		

Figure A.3 – Possible arrester locations in symmetrical monopole VSC converter stations



Key

A:	AC side arrester	LV:	Reactor valve side arrester (bridge arm)
AV:	Connecting transformer valve side arrester	DB:	DC bus arrester
AR:	Arrester between reactor terminals(bridge arm)	DL:	DC line arrester
E/E1/E2:	Neutral bus arrester	EL/EM:	Electrode line / Metallic return arrester
NBS:	Neutral bus switch		

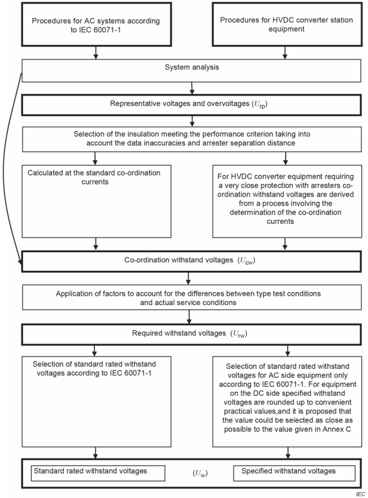
Figure A.2 – Possible arrester locations in one pole of bipolar of VSC converter stations

IEC 60071-11:2022.

Withstand Voltage

Withstand voltage definition process

1. Determination of the representative overvoltages U_{rp}
2. Determination of the co-ordination withstand voltages U_{cw}.
   For HVDC it can be assumed:
   $$U_{cw} = U_{rp}$$
3. Determination of the required withstand voltages U_{rw}
4. Determination of the specified withstand voltages U_w



IEC 60071-11:2022.

Representative overvoltages (U_{rp})

IEC 60071-11:2022

"... representative overvoltages are equal to protection levels of arresters for directly protected equipment."

- The statement is misleading: the arrester protection level is defined by the arrester design and the protection-coordination current amplitude and shape (lightning, switching).
- Representative overvoltages can be determined by assuming maximum system currents and switching impulse, or by transient-simulation analysis of relevant faults.
- Calculation and simulation results without any margin are considered as the representative overvoltage U_{rp}.

Required withstand voltages (U_{rw})

U_{rw} is defined as:

$$U_{rw} = K_a \cdot K_s \cdot U_{rp}$$

K_a Ambient condition correction factor

- Altitude correction
- Temperature / humidity correction for indoor installations

K_s Safety factor

- Altitude correction up to 1000m
- allowance for ageing of insulation
- allowance for changes in arrester characteristics
- allowance for dispresion in the product quality

Required withstand voltages (U_{rw})

Definition safety factor K_s

Table 3 – Indicative values of ratios of required impulse withstand voltage to impulse protective level

Type of equipment	Indicative values of required impulse withstand voltage/impulse protective level[a, c]		
	RSIWV/SIPL	RLIWV/LIPL	RSFIWV/STIPL[b]
AC switchyard – busbars, outdoor insulators, and other conventional equipment	1,20	1,25	1,25
AC filter components	1,15	1,25	1,25
Transformers (in oil)			
line side	1,20	1,25	1,25
valve side	1,15	1,20	1,25
Converter valves	1,15	1,15	1,20
DC valve hall equipment	1,15	1,15	1,25
DC switchyard equipment (including DC filters etc. and DC reactor)	1,15	1,20	1,25

[a] Indicated values are stated for general design objectives only. Appropriate final ratios (higher or lower) can be selected according to the chosen performance criteria.

[b] STIPL for LCC valve arresters.

[c] Indicative ratios are on the basis that any equipment is directly protected with a surge arrester.

IEC 60071-11:2022

Specified withstand voltages (U_w)

- Values are equal to or higher than U_{rw}
- For AC equipment they correspond to the standard values in IEC 60071-1
- For HVDC equipment they are rounded up to convenient values (in most cases IEC 60071-1)

Proposed U_w / kV

550	1300
650	1425
750	1550
850	1675
950	1800
1050	1950
1175	

Example for insulation point coordination

Arrester: DC line/cable arrester (DL)

Arrester design:

Parameter	Value
CCOV	$515kV$
Columns	8
Energy capability	$17MJ$

System analysis:

Parameter	Value
SIPL (U_{rp} for SI)	$807kV@1kA$
LIPL (U_{rp} for LI)	$872kV@5kA$

Required Switching/Lightning Impulse withstand voltage (RSIWV/ RLIWV):

Switching: $U_{rw} = 1.15 \cdot 807kV = 928kV$

Lightning: $U_{rw} = 1.2 \cdot 872kV = 1046kV$

Switching/Lightning Impulse withstand voltage (SIWV/ LIWV):

Switching: $U_w = 950kV$

Lightning: $U_w = 1050kV$